| Question | | | Answer | Marks | AOs | Guidance | |
|---|---|---|---|---|---|---|---|
| 10 | (a) | | Acceleration due to gravity is not a constant but depends on location in the universe. | B1<br>[1] | 3.5b | Allow any sensible comment that $g$ might be different on the moon | |
| 10 | (b) | | Using $s = 2, \quad u = 0, \quad t = 1.6$ and $s = ut + \frac{1}{2}at^2$<br>Downwards as the positive direction<br>$2 = \frac{1}{2} \times 1.6^2 a$ giving $[a = 1.5625]$<br>which 1.6 m s$^{-2}$ to 2 sf | M1<br><br>A1<br>[2] | 3.3<br><br>1.1 | Allow any sign convention<br><br>Allow $-1.6$ m s$^{-2}$ if upwards is clearly indicated as positive. Must be 2 significant figures. | |
| 10 | (c) | | Using $u = 15, \quad v = 0, \quad a = -1.6$ and $v^2 = u^2 + 2as$<br>Upwards as the positive direction<br>$0 = 15^2 - 2 \times 1.6s$<br><br>$s = 70.3$ m | M1<br><br>A1<br>[2] | 3.4<br><br>1.1 | Use of *suvat* equation(s) leading to a value for $s$. Allow sign errors<br>Must follow from correct working and if negative, explained. | Allow answers in the range 70 to 72.2 |